18-874: Spacecraft Design-Build-Fly Lab

Spring 2026

Course Description

Spacecraft design is a truly interdisciplinary subject that draws from every branch of engineering. This capstone design class brings together the material from prior classes in a way that emphasizes the interactions between disciplines and demonstrates how some of the more theoretical topics are synthesized in the practical design of a spacecraft. The class will design, build, and test a small satellite that addresses objectives and requirements posed at the beginning of the course sequence. Students will work in subsystem teams, each focusing on some aspect of the spacecraft, but exposed to many different disciplines and challenges. Practical, hands-on engineering skills will be emphasized, along with fabrication and testing of physical hardware and the creation of thorough documentation.

Special note for Spring 2026:

This offering of the course is different from all prior offerings of the course in that it is being offered for a single semester, not for the typical two-semester design-build-fly cycle. In this case, the teams in the course will be taking custody of engineering tasks leading up to the launch of two Argus-2 satellites, with a launch date of June 2026, Integration date of April 17 2026, and Testing deadline of April 1 2026. In parallel with launch readiness for Argus-2, the team will be developing and implementing features for Argus-3, which will launch approximately 12-18 months in the future.

Instructors

Prof. Brandon Lucia	**Email:** blucia@cmu.edu
System Scientist & Labmaster: Neil Khera	**Email:** nkhera@andrew.cmu.edu

Learning Objectives

The goal of this course is to give students hands-on experience designing and building small spacecraft subsystems and integrating them into a CubeSat. Throughout this course, students will:

1. Understand how the design and integration of a system whose performance depends on the success of many interacting subsystems.
2. Work within a small team to fabricate, and test hardware and software through rapid design iteration.
3. Coordinate with other teams to integrate subsystems into a complete spacecraft.
4. Gain exposure to the complete life cycle of a small satellite mission.

5. Develop skills at project management and systems engineering
6. Develop social skills and planning skills to work with a large group and coordinate many required tasks to launch-ready a satellite.

Logistics

The course will involve designing and building hardware in small teams. Class time will be used primarily for weekly team meetings and consulting time to meet with the instructors.

- All-hands meetings will be held at 9:30 on Wednesdays, followed by consulting hours.
- Mandatory lab working sessions will be held on 9:30 on Mondays.
- Sub team meetings will be held once per week at times coordinated with the instructors.
- Attendance of weekly team meetings is mandatory.
- Slack will be used for coordination between teams and instructors. All students will be added to the "SpacecraftDesignBuildFlyLab" slack channel.
- GitHub will be used to manage project files for all teams and teams will be required to use good git hygiene and software engineering practices.

Assignments and Exams

There will be no exams in this course. Evaluation will be based on participation, contribution to design and fabrication work, final documentation, and weekly project management metrics from each team.

Grading

Grading will be based on:

- 20% Participation and attendance of team meetings
- 30% Individual technical contributions quantified by git commit history / number and significance of PRs and peer surveys
- 30% Completeness and quality of team documentation
- 20% Final presentation and report on team progress for semester

Learning Resources

There is no textbook required for this course.

Course Policies

Attendance: This is a team-based course. In order to coordinate work among teams, participation in weekly meetings is required. If you are unable to be present at a meeting, you must notify the instructors and ensure that your teammates are prepared to present your work.

Accommodations for Students with Disabilities: If you have a disability and are registered with the Office of Disability Resources, I encourage you to use their online system to notify me of your accommodations and discuss your needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu.

Statement of Support for Students' Health & Well-Being: Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep, and taking some time to relax. This will help you achieve your goals and cope with stress.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit http://www.cmu.edu/counseling. Consider reaching out to a friend, faculty, or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:
CaPS: 412-268-2922
Re:solve Crisis Network: 888-796-8226

If the situation is life threatening, call the police:
On campus: CMU Police: 412-268-2323
Off campus: 911

Tentative Spring 2026 Schedule

Week	Dates	Topics
1	Jan 12	Course Overview & Logistics
	Jan 14	Mission PDR Review and Team Assignments
2	Jan 19	No Class (MLK Day)
	Jan 21	Argus-3 Objectives
3	Jan 26	Project Management
	Jan 28	All-Hands Meeting
4	Feb 2	Lecture Orbital Edge Computing (In Lab)
	Feb 4	All-Hands Meeting
5	Feb 9	Lab Day (Prof. Lucia Out)
	Feb 11	Lab Day (Prof. Lucia Out)
6	Feb 16	Lecture Efficient's Dataflow Architecture
	Feb 18	All-Hands Meeting
7	Feb 23	Lecture Embedded Rust
	Feb 25	All-Hands Meeting
8	Mar 2	No Class (Spring Break)
	Mar 4	No Class (Spring Break)
9	Mar 9	Case Study (short lecture): Tartan Artibeus
	Mar 11	All-Hands Meeting
10	Mar 16	Guest Lecture or Lab Day
	Mar 18	All-Hands Meeting
11	Mar 23	Lab Work Day
	Mar 25	No Class (ASPLOS)
12	Mar 30	Guest Lecture or Lab Day
	Apr 1	All-Hands Meeting
13	Apr 6	Guest Lecture or Lab Day
	Apr 8	All-Hands Meeting
14	Apr 14	Lab Work Day
	Apr 16	All-Hands Meeting
14	Apr 21	Final Presentations / Design Review
	Apr 24	Final Presentations / Design Review